Sr.No. 8516

Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

AY: 2025-26
Session: Odd MSE October 2025
Date: 08-10-2025
Sem: 1
Slot: 10:00am - 11:00am
C. Code: CS413
Exam Type: Written
Block:401
Course Section:1
Seat No: 2022600025
1_29_482_986_1_1505_38452

Name of the candidate. Ruchir Khare

Candidate's UID number: 2 0 2 2 6 0 0 0 2 5

Examination class room number: 4 0 1

Invigilator's signature with date: 8/10/25

(To be filled in by the candidate)

EXAMINATION: BE BTECH
(For example FY MCA/FY BTECH)

SEMESTER : 7

PROGRAM: AIML

~~I / II / III / IV / V / VI~~ / VII / ~~VIII~~

DAY AND DATE: 8/10/25 Wednesday

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

a128

INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.

2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.

3) Do not write anything in the column provided for the marks to be assigned by the examiners.

4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.

5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.

6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.

7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.

8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**

9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.

10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.

11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.

12) The answer-books will be scrutinized before they are sent to examiners.

13) All answer-books supplied shall be returned whether written or blank.

14) Nothing shall be written on the question-paper.

15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.

16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.

17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Space For Marks	Question No.	
	Q1 a)	

$$f(w) = \frac{1}{2} \|Xw - y\|^2$$

gradient update rule :

$$w_{new} = w_{old} - \frac{\partial L(\theta)}{\partial w}$$

where $L(\theta)$ is the loss fn which is to be minimized

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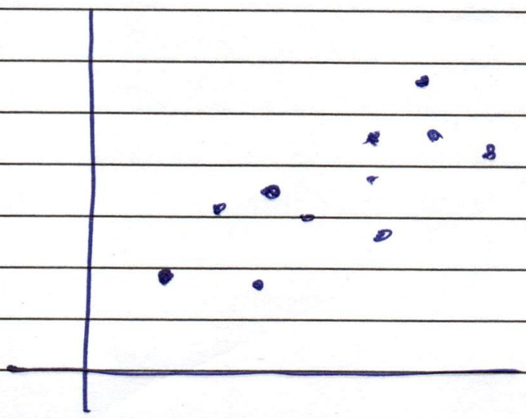
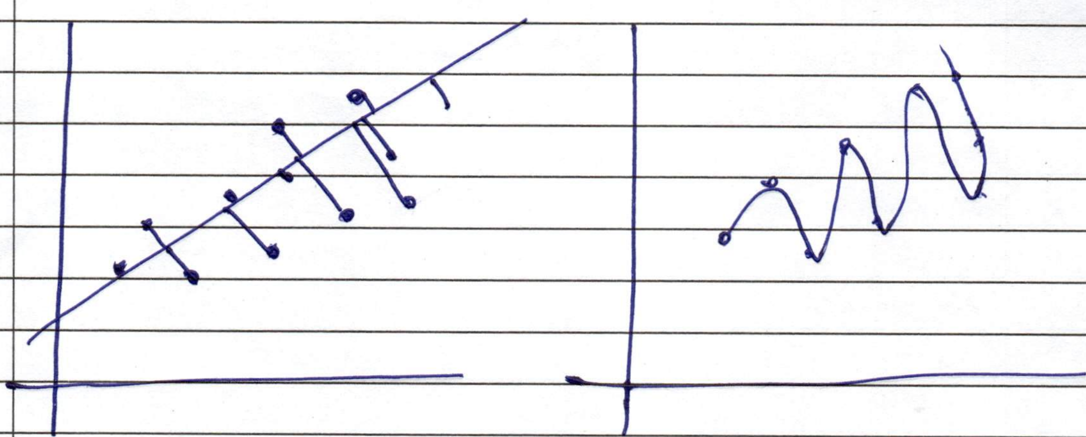
Q1 b) The square error loss function is the most common form of a loss function where the main goal is to minimize the square error loss. so as to update the weights and biases at every epoch of the ~~function~~ algorithm.

- In general for a machine learning model to generalise well and perform accurately, a low bias and low variance is necessary.

- 

for the above sample a good model would make fit a linear line that would provide some error in train and test both but it generalises well



~~good~~ fairly accurate model

overfit model (high bias)

~~This show~~

- When the model capacity is too high i.e. when it starts to provide very high training accuracy, its safe to say that the model has overfitted.

- This is because it means that model has also learned

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about the errors of the given dataset

- This is not very good in general for the ML algorithm because now the model would not be able to ~~perform~~ generalise well to unseen data.

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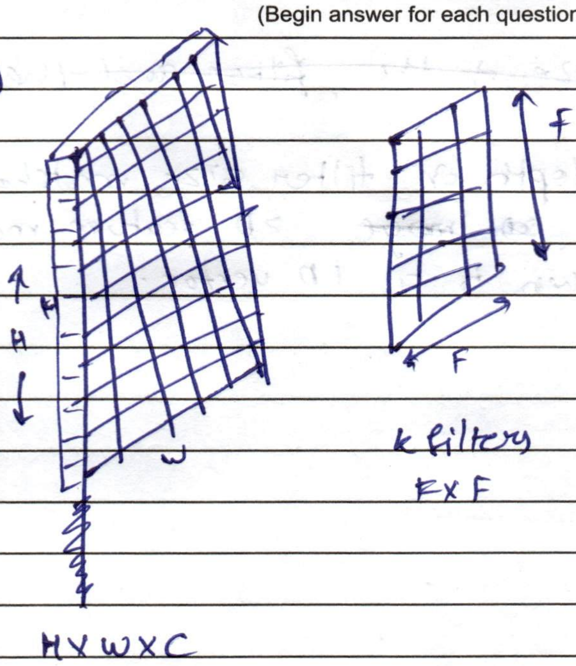
Q2a)



~~The size~~

- When an image ~~is~~ of size $H_1 \times W_1 \times C$, has a filter of F×F applied ~~to~~ the size of the output image is

$$H_2 = \frac{H_1 - F + 2P}{S} + 1$$

where $S \rightarrow$ stride
$P \rightarrow$ padding.

$$W_2 = \frac{W_1 - F + 2P}{S} + 1$$

- ~~This~~ Thus to apply a single convolution operation we maybe having to have atleast F×F operations.

- Now, because there are k such filters, the ~~size the~~ operation size increases to $\rightarrow \boxed{K \times F \times F}$,

- Usually when filters are applied they are of same depth as the image, but since here we have a 2D filter we need C times more ~~p~~ operations hence the total operations ~~so~~ would be approximately $\boxed{K \times F \times F \times C}$

- Depth wise seperable filters are very common in CNN ~~no~~ arch. because at the end of the CNN layers we have to stretch the 2D image ~~into~~ into 1D and then ~~@~~ provide that to the fully connected network.

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- ~~Thus when the size of the filter and the~~

- Thus when the depth of filter size matches the image we get a ~~2D image~~ 2D feature map, which is easy to ~~st~~ flatten to a 1D vector.

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Q2 b) ~~A linear~~

- An autoencoder can function ~~as a~~ similar to PCA only when
  1) Input and output activation functions are linear.
  2) $dim(X) > dim(H)$
  3) loss function employed is square error.
  4) The input ~~is~~ ~~covered~~ is applied with the following function

~~The MSE loss~~ $f(X_t) = \dfrac{1}{\sqrt{m}} \dfrac{\sum_{n=1}^{n} m}{n}$

The mean square error loss function tries to minimise

$$\mathcal{L}(\theta) = \frac{1}{n} \sum |y - \hat{y}|^2$$

The formula for gradient descent is

$$w_{new} = w_{old} - \frac{\partial \mathcal{L}(\theta)}{\partial w}$$

$$\frac{\partial \mathcal{L}(\theta)}{\partial w} = 2|y - \hat{y}|$$

$\hat{y} = wx + b$

Therefore putting back in

$$\frac{\partial \mathcal{L}(\theta)}{\partial w} = x^T x - wx$$

This is exactly ~~what~~ ~~PCA~~ similar to the ~~covarian~~ covariance matrix that PCA uses. thus we can say that both ~~PCA~~ ~~and~~ PCA and undercomplete autoencoder perform similar tasks.

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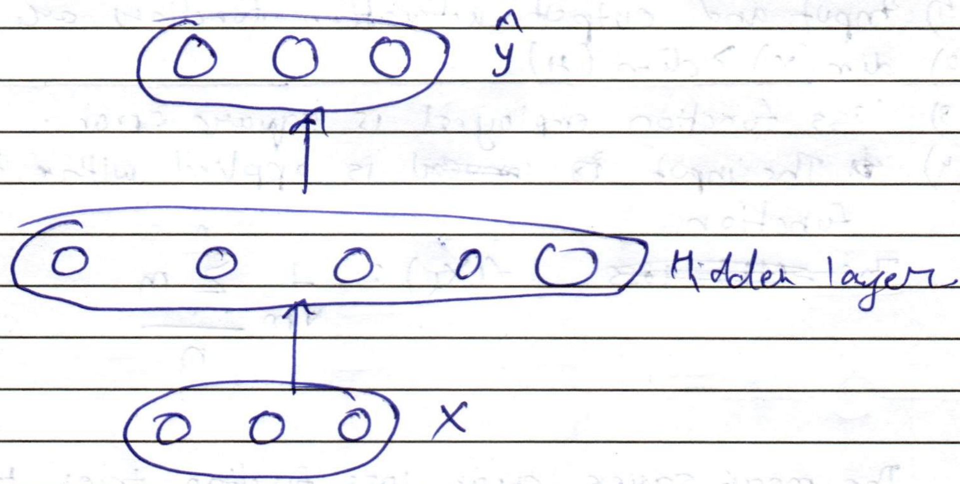
An overcomplete autoencoder is the one in which dim(hidden) $\geq$ dim(i/p)



very intuitively from the architecture we can infer that there is a very high tendency for the hidden layer to just copy the inputs and reproduce it as it is.

$H = wX + b$

when we apply regularisation this does not happen because now there is another ~~term~~ parameter.

~~$WX + b + ||T||$~~ $wX + b + ||T||$

Now this new function effectively penalises the Autoencoder for learning the input as it is. This helps in enhanced feature extraction and prevents overfitting.

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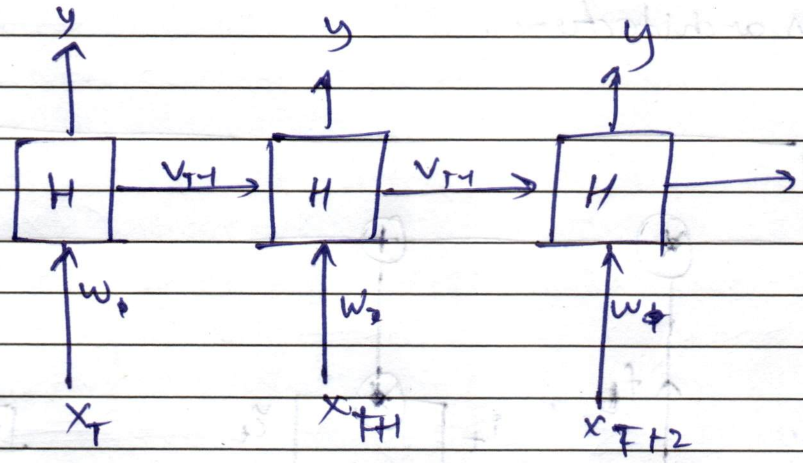
	Q3 a)	



unfolded RNN in time.

- It is evident from diagram that
  $H = WX_T + WV_{T-1} + b$

- This means that a single neurons output ~~is~~ depends on ~~the~~ its previous ones

- while ~~apply~~ gradient descent in backpropagation

$$H_T = H_{T-1} - \frac{\partial L}{\partial h_{t-k}}$$

Thus $\frac{\partial L}{\partial h_{T-K}} = \frac{\partial L}{\partial h_{t-1}} \frac{\partial h_{t-1}}{\partial h_{t-2}} \frac{\partial h_{t-2}}{\partial h_{t-3}} \cdots \frac{\partial h_0}{\partial W}$

By chain rule.

- This is not very good because ~~to~~ towards the end the gradients ~~become~~ tends to zero this makes the RNN have a very ~~short~~ short memory.

- LSTM improves this by having 3 gated architecture
  1) Forget gate
  2) Input gate
  3) Output gate.

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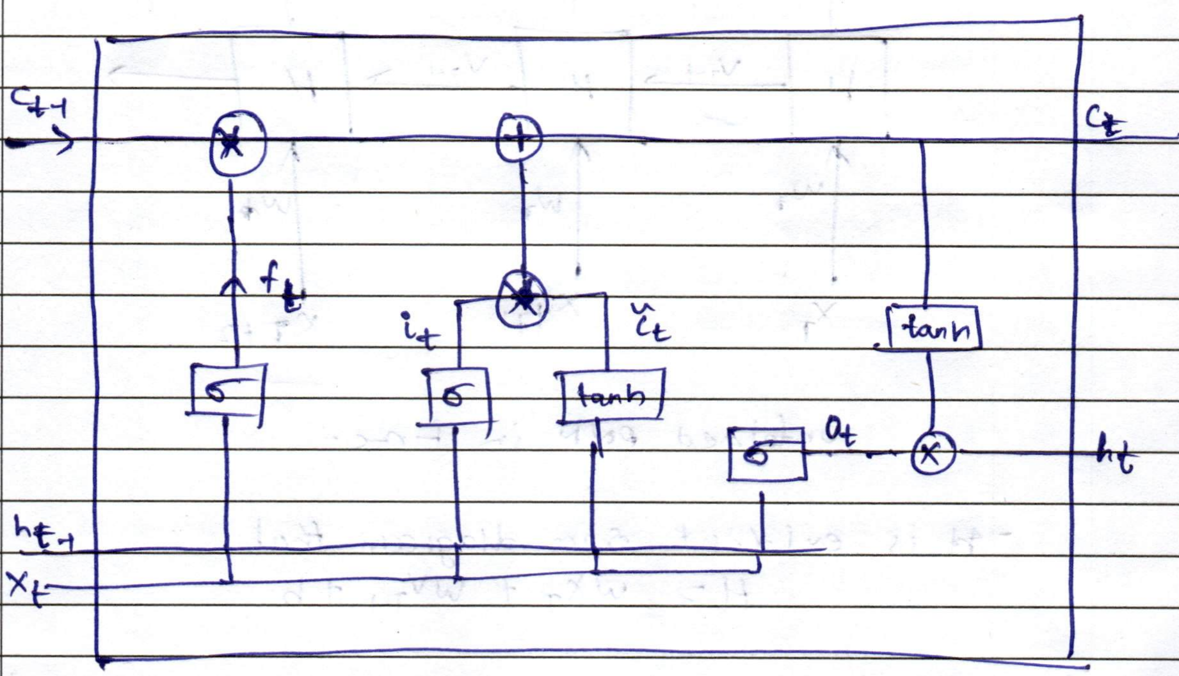
LSTM architecture.



LSTM ~~prevents~~ has a gate dedicated for each of the task needed to perform.

$$f_t = \sigma(W_f x_t + W_f h_{t-1} + b_f)$$

Depending on the previous hidden state, the forget gat actually decides what information to keep

$h_{t-1} = [0.5 \quad 0.2 \quad 0.1 \quad 0.9]$

$x_t = [0.9 \quad 0.1 \quad 0.5 \quad 0.4]$

In the output ~~was the~~ becase h_{t-1} has a low value for the 3rd parameter it will decide to forget that because the information was not important before so will not be now.

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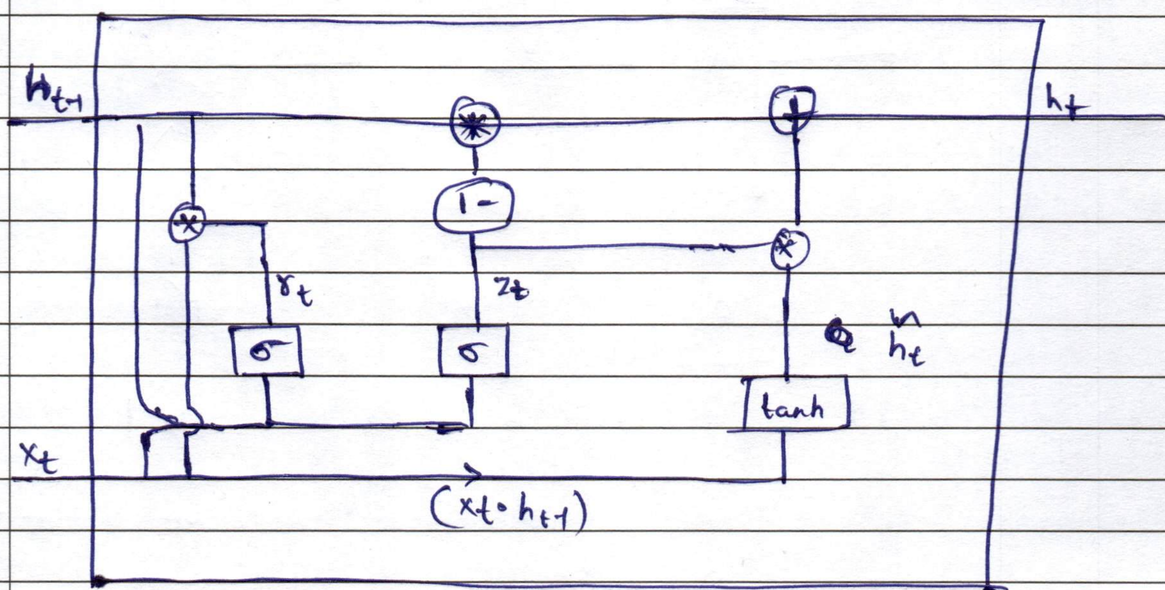
Q3 b) GRU (Gated Recurrent Unit)



$$r_t = \sigma(W_r x_t + W_r h_{t-1} + b_r)$$

$$z_t = \sigma(W_z x_t + W_z h_{t-1} + b_z)$$

$$\hat{h}_t = \tanh(W_c(x_t \cdot h_{t-1}) + W_c x_t + b_c)$$

(candidate state)

This the proof that candidate hidde state is a convex combination of previous hidden state }

$$h_t = ((1 - z_t) \cdot h_{t-1}) + (\hat{h}_t \cdot z_t)$$

- GRU only has 2 gates update gate and reset gate whereas LSTM have 3 gates
- Lessor the gates lesser the trainable parametere
- This enables GRUs to actually be faster in training than LSTM.

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